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Subject: **CS 268**

Aim:

Design and implementation of a pipeline architecture in verilog.

Theory:

Pipelining is a technique used in modern processors to improve performance by executing multiple instructions simultaneously. It breaks down the execution of instructions into several stages, where each stage completes a part of the instruction. These stages can overlap, allowing the processor to work on different instructions at various stages of completion, similar to an assembly line in manufacturing.

Verilog Code:

Pipeline.v:

```
module pipe_ex (F, A, B, C, D, clk);  
    parameter N = 10;  
  
    input [N-1:0] A, B, C, D;  
    input clk;  
    output [N-1:0] F;  
  
    reg [N-1:0] L12_x1, L12_x2, L12_D;  
    reg [N-1:0] L23_x3, L23_D;  
    reg [N-1:0] L34_x4, L34_D;  
    reg [N-1:0] L45_F;
```

```

assign F = L45_F;

always @(posedge clk)
begin
    L12_x1 <= #4 A + B;
    L12_x2 <= #4 C - D;

    L12_D <= D; // ** STAGE 1 **

    L23_x3 <= #4 L12_x1 + L12_x2;
    L23_D <= L12_D; // ** STAGE 2 **

    L34_x4 <= #4 L23_x3 * L23_D;
    L34_D <= L23_D; // ** STAGE 3 **

    L45_F <= #4 L34_x4 + L34_D; // ** STAGE 4 **
end

endmodule

```

Pipeline tb.v:

```

`timescale 1ns/1ps

module pipeline_tb;

parameter N = 10;

reg [N-1:0] A, B, C, D;

reg clk;

wire [N-1:0] F;

// Instantiate the pipeline module
pipeline uut (.F(F), .A(A), .B(B), .C(C), .D(D), .clk(clk));

```

```

// Clock generation
always #5 clk = ~clk; // 10ns clock period

initial
begin
    // Initialize inputs
    clk = 0;
    A = 10; B = 5; C = 20; D = 2;

    // Apply test cases
    #10 A = 15; B = 7; C = 30; D = 4;
    #10 A = 8; B = 3; C = 10; D = 1;
    #10 A = 12; B = 6; C = 25; D = 5;
    #20 $stop; // End simulation
end

// Monitor output
initial
    $monitor("Time=%0t | A=%d B=%d C=%d D=%d | F=%d", $time, A, B, C,
D, F);

endmodule

```

Output:

The screenshot displays the Vivado IDE's logic analyzer interface. On the left, a list of signals is shown with their current values. On the right, a timing diagram visualizes the digital signals over a 50 ns period. The signals A, B, C, and D are represented by green waveforms with hex values. Signal F is a red waveform labeled 'xxx'. Signal N is a green waveform labeled '0000000a'. The clk signal is a green square wave.

Signal	Value
A[9:0]	00c
B[9:0]	006
C[9:0]	019
D[9:0]	005
clk	1
F[9:0]	0c0
N[31:0]	0000000a